Max Circular Subarray Sum

Difficulty: Hard | Accuracy: 29.37% | Submissions: 208K+ | Points: 8 | Average Time: 25m

You are given a circular array `arr[]` of integers, find the maximum possible sum of a non-empty subarray. In a circular array, the subarray can start at the end and wrap around to the beginning. Return the maximum non-empty subarray sum, considering both non-wrapping and wrapping cases.

Examples

Input: arr[] = [8, -8, 9, -9, 10, -11, 12]

Output: 22

Explanation: Starting from the last element of the array, i.e, 12, and moving in a circular fashion, we have max subarray as 12, 8, -8, 9, -9, 10, which gives maximum sum as 22.

Input: arr[] = [10, -3, -4, 7, 6, 5, -4, -1]

Output: 23

Explanation: Maximum sum of the circular subarray is 23. The subarray is [7, 6, 5, -4, -1, 10].

Input: arr[] = [5, -2, 3, 4]

Output: 12

Explanation: The circular subarray [3, 4, 5] gives the maximum sum of 12.

Constraints

$1 \le arr.size() \le 10^5$

$-10^4 \le arr[i] \le 10^4$

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